Academic case study | Sophia Learning — Introduction to Networking
GreenLeaf Network Infrastructure & Core Services

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Date: 09/10/2026

1. Network Topology

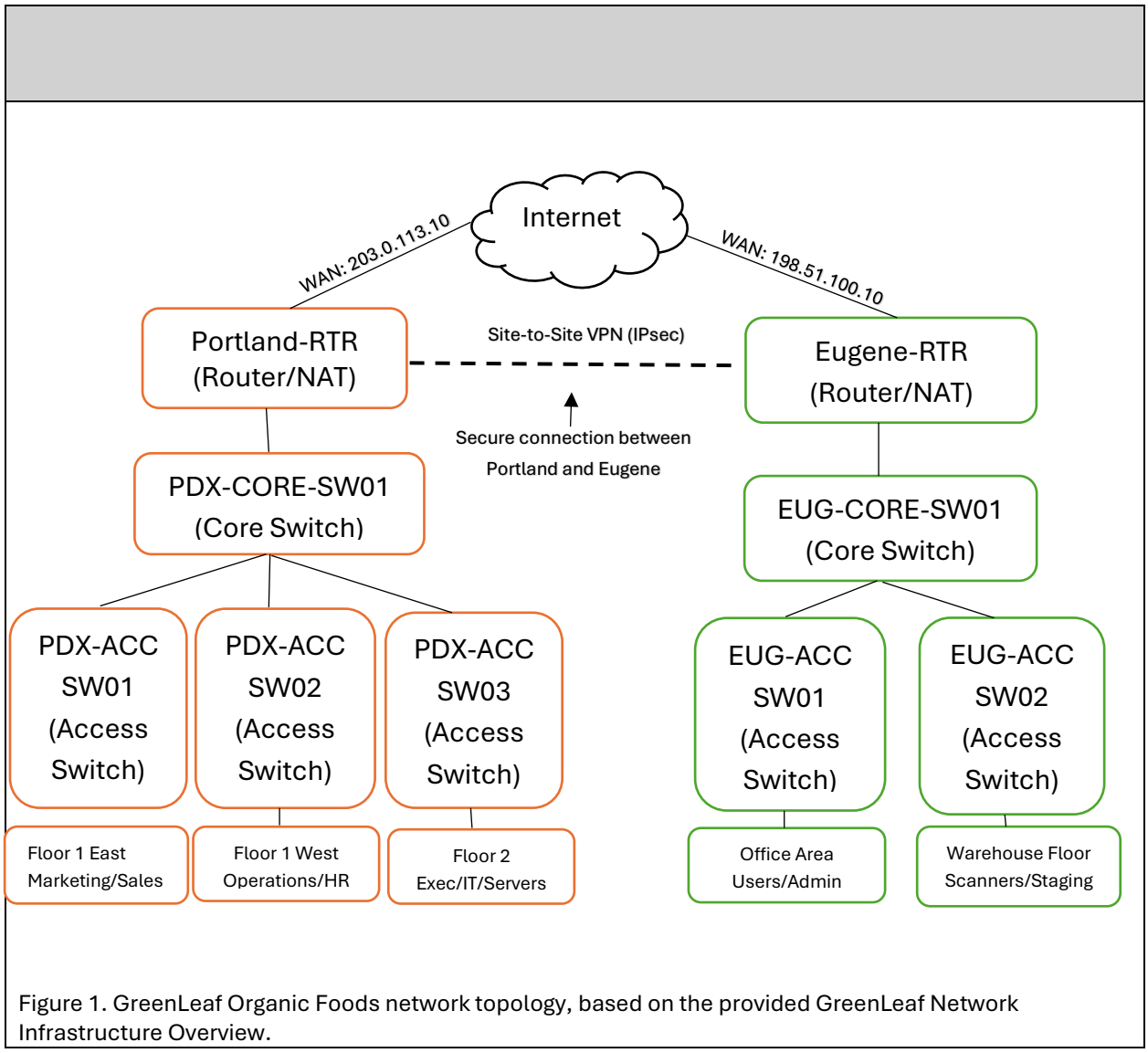


Figure 1. GreenLeaf Organic Foods network topology, based on the provided GreenLeaf Network Infrastructure Overview.

2. Protocols and Encapsulation

2a. Protocols Observed in Packet Captures

Table 1. Protocols Observed in Packet Captures

Protocol	Where It Appears	Purpose in the Network
TCP	Capture 1 – TCP Three-Way Handshake	TCP is used to create a reliable connection between devices before they start exchanging data. In this capture, the workstation connects to the ERP server using SYN, SYN-ACK, and ACK process.
DHCP	Capture 2 – DHCP Discovery Process	DHCP automatically gives devices the network information they need to connect. In this capture, the client goes through the Discover, Offer, Request, and ACK process to receive its IP address and other network settings.
DNS	Capture 3 – DNS Query and Response	DNS helps devices find other systems by translating a hostname into an IP address. In this capture, the workstation looks up pdx-erp01.greenleaf.local, and the DNS server responds with the IP address 10.10.30.20.

2b. Encapsulation Summary

When a user clicks “View Inventory” in the ERP application, the request moves through Layers 7 through 4, where the data is encrypted with TLS and TCP adds the information needed for the connection. At Layer 3, the source and destination IP addresses are added, while Layer 2 adds the MAC addresses to prepare the data to be sent across the network. Finally, Layer 1 turns everything into electrical signals and sends it through the network cable.

3. IP Addressing and Subnetting

3a. Portland Subnets

Table 2. Portland Subnets

VLAN	Network	Mask/CIDR	First Usable	Last Usable	Broadcast	Gateway
1	10.10.1.0	255.255.255.0/24	10.10.1.1	10.10.1.254	10.10.1.255	10.10.1.1
10	10.10.10.0	255.255.255.0/24	10.10.10.1	10.10.10.254	10.10.10.255	10.10.10.1
20	10.10.20.0	255.255.255.0/24	10.10.20.1	10.10.20.254	10.10.20.255	10.10.20.1
30	10.10.30.0	255.255.255.128/25	10.10.30.1	10.10.30.126	10.10.30.127	10.10.30.1
40	10.10.40.0	255.255.255.0/24	10.10.40.1	10.10.40.254	10.10.40.255	10.10.40.1
50	10.10.50.0	255.255.255.0/24	10.10.50.1	10.10.50.254	10.10.50.255	10.10.50.1

3b. Eugene Subnets

Table 3. Eugene Subnets

VLAN	Network	Mask/CIDR	First Usable	Last Usable	Broadcast	Gateway
1	10.20.1.0	255.255.255.0/24	10.20.1.1	10.20.1.254	10.20.1.255	10.20.1.1
10	10.20.10.0	255.255.255.128/25	10.20.10.1	10.20.10.126	10.20.10.127	10.20.10.1
40	10.20.40.0	255.255.255.128/25	10.20.40.1	10.20.40.126	10.20.40.127	10.20.40.1
50	10.20.50.0	255.255.255.128/25	10.20.50.1	10.20.50.126	10.20.50.127	10.20.50.1
30	10.20.30.0	255.255.255.128/25	10.20.30.1	10.20.30.126	10.20.30.127	10.20.30.1

3c. Subnet Calculation (Show Your Work)

Subnet: **Eugene VLAN 10 – 10.20.10.0/25**

A /25 subnet uses 25 of the 32 bits for the network, which leaves 32 – 25 = 7 bits for the hosts. To find the total number of addresses, I calculated 2 to the 7th power which equals 128 total addresses. Two addresses cannot be used by devices because one is the network address and one is the broadcast address, so after subtracting those two, 128 – 2 = 126 usable hosts.

Network Address: 10.20.10.0
First Usable IP: 10.20.10.1
Last Usable IP: 10.20.10.126
Broadcast Address: 10.20.10.127
Usable Hosts: 126

IPv6: IPv6 is partially deployed at the Portland headquarters and is currently used on two VLANs. VLAN 10 (Users) uses the 2001:db8:10:10::/64 prefix with SLAAC, while VLAN 30 (Servers) uses the 2001:db8:10:30::/64 prefix with static addressing. IPv6 is not currently used at the Eugene location.

4. Core Network Services

4a. DHCP

DHCP is provided by PDX-DC01 at 10.10.30.10. In Portland, DHCP is used for VLAN 10 (Users), VLAN 20 (VoIP), VLAN 40 (Wireless), and VLAN 50 (Guest). In Eugene, DHCP is used for VLAN 10 (Users), VLAN 40 (Wireless), and VLAN 50 (Warehouse), with the Eugene router relaying DHCP requests to the Portland server. Lease times vary depending on the type of device, including 8 hours for most users, 4 hours for wireless and warehouse devices, and 2 hours for Portland guest devices. DHCP also provides the default gateway, DNS servers addresses, and the greenleaf.local domain name to clients.

4b. NAT/PAT

Both Portland and Eugene routers use GigabitEthernet0/0/0 as the NAT outside interface because it connects to the WAN. In Portland, VLANs 10, 20, 30, 40, and 50 are configured as NAT inside, while Eugene uses VLANs 10, 30, 40, and 50 as NAT inside. Both routers use PAT with overload, which allows multiple devices on the internal networks to share the router's public IP address when accessing outside networks. The Portland NAT ACL allows the 10.10.10.0/24, 10.10.20.0/24, 10.10.30.0/25, 10.10.40.0/24, and 10.10.50.0/24 networks. The Eugene NAT ACL allows the 10.20.10.0/25, 10.20.30.0/25, 10.20.40.0/25, and 10.20.50.0/25 networks.

4c. DNS Naming

DNS is provided by PDX-DC01 at 10.10.30.10 and PDX-DC02 at 10.10.30.11. Devices use the greenleaf.local domain, so hostnames are formed by adding the device name before the domain, such as pdx-erp01.greenleaf.local. Dynamic DNS updates are enabled and set to secure only, which means updates require Active Directory authentication.